

# FINAL EXPLANATION: CORE MATHEMATICS GRADE 10

*Teacher Key: Exemplar Answers and Marking Guide*

## FINAL EXPLANATION: CORE MATHEMATICS GRADE 10 – SUB-STRAND 2.4 TRIGONOMETRY 1: FINAL EXPLANATION

Teacher Key: Exemplar Answers (do not give to students)

### INSTRUCTIONS GIVEN TO STUDENTS

Driving question: How can we use the angles and one side of a right-angled triangle to find any unknown height or distance, and how does this help Kenyans solve real problems in construction, surveying and everyday life? Answer every section in the space below it. Use the Kenya Power mast near the school in Nakuru County as your main example: a student stands 40 m from the base of the mast and measures the angle of elevation to the top as about  $50^\circ$ . Draw neat, labelled diagrams, show every step of your working, include units, and write a full sentence for each final answer. You may use your mathematical tables, calculator and Special Angles Reference Card. Where a table value is needed it is given to you.

### Part 1: The missing tool

**On Day 1 you were asked whether you could find the height of the Nakuru mast using only a horizontal distance of 40 m and an angle of about  $50^\circ$ . (a) Draw the right-angled triangle for this situation. Mark the  $50^\circ$  angle and the 40 m side, and label the three sides as opposite, adjacent and hypotenuse relative to the  $50^\circ$  angle. (b) Explain why Pythagoras' theorem alone could not solve the problem. (c) In Lesson 1 you drew right-angled triangles of different sizes that all shared the same acute angle and calculated side ratios. What did you notice about the ratios, and why does this make a table of ratios useful for finding the mast's height? (d) Write the names and definitions (as ratios of sides) of sine, cosine and tangent.**

(a) A right-angled triangle with the horizontal ground line of 40 m from the base of the mast to the student, the vertical mast as the upright side, and the line of sight from the student to the top as the longest side. The  $50^\circ$  angle is at the student's position. Relative to the  $50^\circ$  angle: the mast height ( $h$ ) is the opposite side, the 40 m ground distance is the adjacent side, and the line of sight is the hypotenuse. (b) Pythagoras' theorem connects the lengths of two sides to find a third side, but here only one side (40 m) is known, together with an angle. Pythagoras has no way to use an angle, so we need a different tool that links an angle to a ratio of sides. (c) In triangles of different sizes with the same acute angle, the ratio of any chosen pair of sides stayed nearly constant. This means the ratio depends only on the angle and not on the size of the triangle. So if the ratio for  $50^\circ$  is known from a table, it can be used for the huge mast triangle, even though we could only measure small triangles on paper. (d) Sine of an angle = opposite  $\div$  hypotenuse ( $\sin \theta = O/H$ , SOH). Cosine of an angle = adjacent  $\div$  hypotenuse ( $\cos \theta = A/H$ , CAH). Tangent of an angle = opposite  $\div$  adjacent ( $\tan \theta = O/A$ , TOA).

## Part 2: Solving the mast problem

Use the triangle from Part 1. (a) Which trigonometric ratio links the  $50^\circ$  angle, the 40 m side and the unknown height  $h$ ? Give a reason based on which sides are involved. (b) Write the equation, rearrange it to make  $h$  the subject, and calculate the height of the mast. Use  $\tan 50^\circ \approx 1.192$ . (c) Explain in a sentence how this answers the question from Day 1 without using ladders, drones or guessing.

(a) The tangent ratio. The height  $h$  is the opposite side and the 40 m distance is the adjacent side, and  $\tan \theta = \text{opposite} \div \text{adjacent}$  uses exactly these two sides and the angle. (b)  $\tan 50^\circ = h \div 40$ . So  $h = 40 \times \tan 50^\circ = 40 \times 1.192 = 47.68 \text{ m} \approx 47.7 \text{ m}$ . The mast is about 47.7 m tall. (c) By measuring only the horizontal distance (40 m) and the angle of elevation ( $50^\circ$ ) from the ground, the tangent ratio gives us the height directly, so no one needs to climb the mast. The tangent ratio is the mathematical bridge between the known side and the unknown height.

## Part 3: Relationships between the ratios

(a) In the mast triangle, work out the size of the other acute angle (at the top of the mast). Then compare  $\sin$  of that angle with  $\cos 50^\circ$ . Use the triangle to explain what you find. Your explanation must refer to opposite and adjacent sides. State the general rule for any acute angle  $\theta$ . (b) Suppose the  $\tan$  button on a surveyor's calculator is broken. Which identity connecting  $\tan \theta$  with  $\sin \theta$  and  $\cos \theta$  could she use, and why does it work? Show the short algebraic proof using the definitions of the ratios. (c) Given that  $\sin \theta = 3/5$  in a right-angled triangle, find  $\cos \theta$  and  $\tan \theta$  without using a calculator.

(a) The angles of a triangle add to  $180^\circ$  and the right angle is  $90^\circ$ , so the other acute angle is  $90^\circ - 50^\circ = 40^\circ$ . The side opposite the  $40^\circ$  angle is the mast height  $h$ , and this same side is adjacent to the  $50^\circ$  angle. The hypotenuse is the same for both angles. So  $\sin 40^\circ = \text{opposite} \div \text{hypotenuse} = h \div \text{hypotenuse}$ , and  $\cos 50^\circ = \text{adjacent} \div \text{hypotenuse} = h \div \text{hypotenuse}$ . They are the same ratio, so  $\sin 40^\circ = \cos 50^\circ$  (both are 0.6428 from tables). In general,  $\sin \theta = \cos(90^\circ - \theta)$  and  $\cos \theta = \sin(90^\circ - \theta)$ , because the side opposite one acute angle is adjacent to the other. (b) She can use  $\tan \theta = \sin \theta \div \cos \theta$ , so  $\tan 50^\circ = \sin 50^\circ \div \cos 50^\circ$ . Proof:  $\sin \theta \div \cos \theta = (O/H) \div (A/H) = (O/H) \times (H/A) = O/A = \tan \theta$ . The hypotenuse cancels, leaving opposite  $\div$  adjacent, which is the tangent. (c)  $\sin \theta = 3/5$  means opposite = 3 and hypotenuse = 5 (in proportion). The adjacent side =  $\sqrt{5^2 - 3^2} = \sqrt{16} = 4$ . So  $\cos \theta = 4/5$  and  $\tan \theta = 3/4$ . Check:  $\sin \theta \div \cos \theta = (3/5) \div (4/5) = 3/4$ , which agrees.

## Part 4: Special angles, elevation and depression

(a) Imagine the student, still 40

(a)  $\tan 45^\circ = 1$ . This comes from the isosceles right-angled triangle with both legs 1 unit: opposite = adjacent = 1, so  $\tan 45^\circ = 1 \div 1$

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| <p>m from the base, measured an angle of elevation of exactly <math>45^\circ</math> instead of <math>50^\circ</math>. Using the exact value of <math>\tan 45^\circ</math> from your Special Angles Reference Card, find the height of the mast and explain how you obtained the exact value of <math>\tan 45^\circ</math> from a triangle. (b) Now imagine the angle of elevation was <math>60^\circ</math>. Write the exact height using the exact value of <math>\tan 60^\circ</math>, and give it to one decimal place. (c) With the original <math>50^\circ</math> angle of elevation, a worker stands at the top of the mast and looks down at the student on the ground 40 m from the base. What is the angle of depression of the student from the worker? Explain your reasoning, and say how an angle of depression differs from an angle of elevation.</p> | <p>= 1. Then <math>h = 40 \times \tan 45^\circ = 40 \times 1 = 40</math> m. (b) <math>\tan 60^\circ = \sqrt{3}</math>, from the 30-60-90 triangle formed by bisecting an equilateral triangle of side 2 (opposite = <math>\sqrt{3}</math>, adjacent = 1). So <math>h = 40 \times \sqrt{3} = 40\sqrt{3} \approx 69.3</math> m. (c) The angle of depression is <math>50^\circ</math>. The worker's line of sight to the student is the hypotenuse, and the horizontal line from the worker is parallel to the ground. So the angle of depression from the worker to the student equals the angle of elevation from the student to the worker (alternate angles), which is <math>50^\circ</math>. Note that it is measured from the horizontal line, not from the vertical mast, so it is not the <math>40^\circ</math> angle at the top of the mast, which lies between the mast and the line of sight. The angle of elevation is measured upward from the horizontal to the line of sight, while the angle of depression is measured downward from the horizontal to the line of sight.</p> |
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| Part 5: Using trigonometry in real life across Kenya                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
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| <p>A wheelchair ramp is being built at Kisumu County Referral Hospital. It must rise at an angle of <math>12^\circ</math> to the ground and cover a horizontal distance of 6 m. (a) Draw and label the triangle. (b) Use the SOH-CAH-TOA decision steps to find the length of the ramp's sloping surface. Use <math>\cos 12^\circ \approx 0.9781</math> and give your answer to two decimal places. (c) Describe the general steps you follow to solve any right-angled triangle problem. (d) Name two Kenyan professionals who could use these methods, say what each would measure, and explain how this answers the driving</p> | <p>(a) A right-angled triangle with the <math>12^\circ</math> angle at the bottom of the ramp, the horizontal ground distance of 6 m as the adjacent side, the ramp surface as the hypotenuse (x), and the vertical rise as the opposite side. (b) Known: adjacent = 6 m, unknown: hypotenuse. A and H match CAH, so <math>\cos 12^\circ = 6 \div x</math>. Then <math>x = 6 \div \cos 12^\circ = 6 \div 0.9781 \approx 6.13</math> m. The ramp surface is about 6.13 m long. (c) Draw and label the triangle; identify which sides are opposite, adjacent and hypotenuse relative to the given angle; match the known and unknown sides to SOH, CAH or TOA; write and rearrange the equation to isolate the unknown; calculate using tables or a calculator; interpret the answer in context with units. (d) A Kenya Power engineer can find the height of a transmission mast, as in Nakuru, from the distance and the angle of elevation. A land surveyor in the Rift Valley can find the height of a hill or the width of a river from a measured distance and angle. Both need only one side and one angle of a right-angled triangle, and trigonometric ratios give the unknown height or distance without climbing or crossing, which is exactly how trigonometry helps solve real problems in construction, surveying and everyday life.</p> |

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| Criterion                                                 | Excellent (4)                                                                                                                                                            | Proficient (3)                                                                                               | Developing (1–2)                                                                       |
| Labelling and choosing the correct ratio                  | Draws accurate, fully labelled triangles with opposite, adjacent and hypotenuse correct relative to the given angle, and justifies every ratio choice using SOH-CAH-TOA. | Labels triangles mostly correctly and chooses the right ratio, but the justification is brief or incomplete. | Labels sides incorrectly or does not give a reason for choosing the ratio.             |
| Calculation and use of tables, calculator or exact values | Sets up and rearranges equations correctly, calculates accurately (mast about 47.7 m, ramp about 6.13 m, exact special-angle values) with units and sensible rounding.   | Sets up equations correctly with minor arithmetic, rounding or unit errors.                                  | Equations are set up incorrectly or the calculation is mostly incorrect or incomplete. |
| Explaining relationships between ratios                   | Clearly explains $\sin \theta = \cos(90^\circ - \theta)$ using opposite/adjacent swapping and proves $\tan \theta = \sin \theta \div \cos \theta$ from the definitions.  | States the relationships correctly but explanation or proof is partial.                                      | States the relationships without explanation or with errors.                           |
| Elevation, depression and connection to the phenomenon    | Correctly distinguishes elevation from depression, explains why they are equal, and clearly links all work to how the Nakuru mast height is found without climbing.      | Defines elevation and depression correctly with a weak link to the phenomenon.                               | Confuses elevation and depression or does not connect the work to the mast.            |
| Real-life reasoning and communication                     | Gives a clear general method and relevant Kenyan examples, with well-organised working and full-sentence conclusions.                                                    | Gives a reasonable method and examples, with some gaps in organisation.                                      | Gives little or unclear reference to real-life use and working is disorganised.        |

## ATTRIBUTION AND LICENSING

**Curriculum Alignment:** Structured in pedagogical alignment with the competence-based curriculum format and learning outcomes established by the Kenya Institute of Curriculum Development (KICD). This material is independently developed and is not officially endorsed or published by KICD.

**Content & Pedagogy:** Developed and adapted by the educator team at SeaVuria (registered 501(c)(3) in the United States and registered NGO in Kenya, No. 872-850A-BF11), who designed the pedagogical framework, lesson templates and exemplar lessons on which this material is based.

**Production & Layout:** Educational design and curriculum sequencing by SeaVuria. Technical layout, structural formatting, and digital optimization/implementation executed in partnership with ARES Education (registered 501(c)(3) in the United States and Canada (as Afretech Aid) and registered NGO in Kenya), with lesson drafting and compilation assisted by AI (Claude, Anthropic).

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